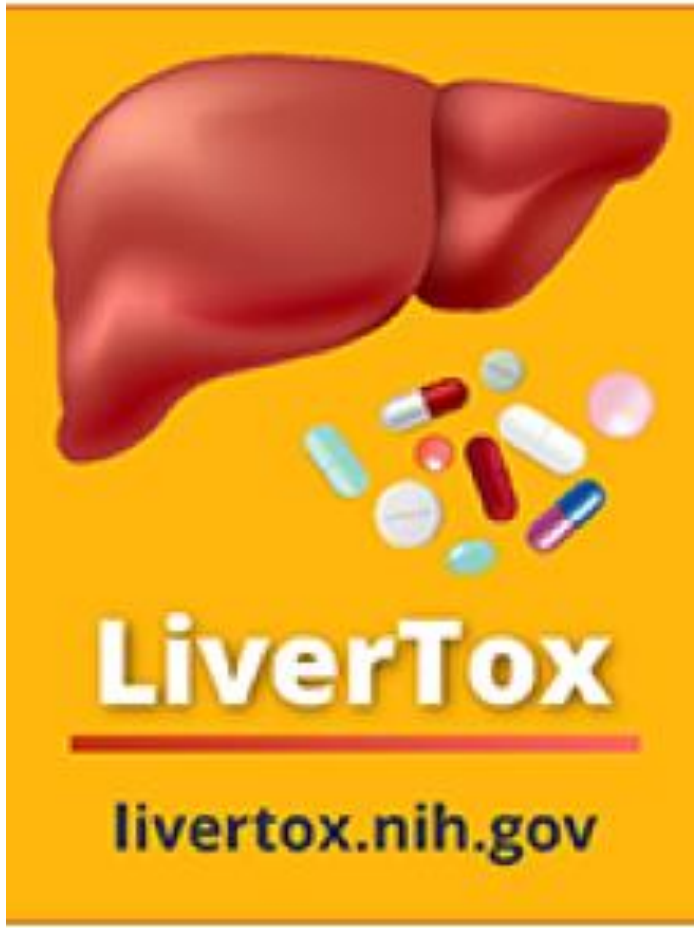
Drug-Induced Liver Injury
DILI

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Drug Induced Liver Injury

- Drug-induced hepatotoxicity or drug-induced liver injury (DILI) is an **acute or chronic response to a natural or manufactured compound.**
- More than 1000 medications and herbal compounds are known to cause hepatotoxicity and can be found on a searchable database maintained by the National Institute of Diabetes and Digestive, and Kidney Diseases (NIDDK) called LiverTox.



- **DILI** is also the most frequently cited reason for withdrawal of medications from the marketplace (up to 32% of drug withdrawals) .

- ***Risk factors:***
- ***Age:*** Adults appear to be at higher risk for DILI than children
- ***Sex:*** Females are more susceptible to DILI
- ***Race and ethnicity***: The predominant drug implicated in DILI in African Americans is trimethoprim-sulfamethoxazole, while in White Americans, it is amoxicillin-clavulanate.
- **Other Factors:** as obesity, diabetes, and chronic viral hepatitis.
- ***Genetic polymorphisms:*** in the CYP isoenzymes

DILI, classification

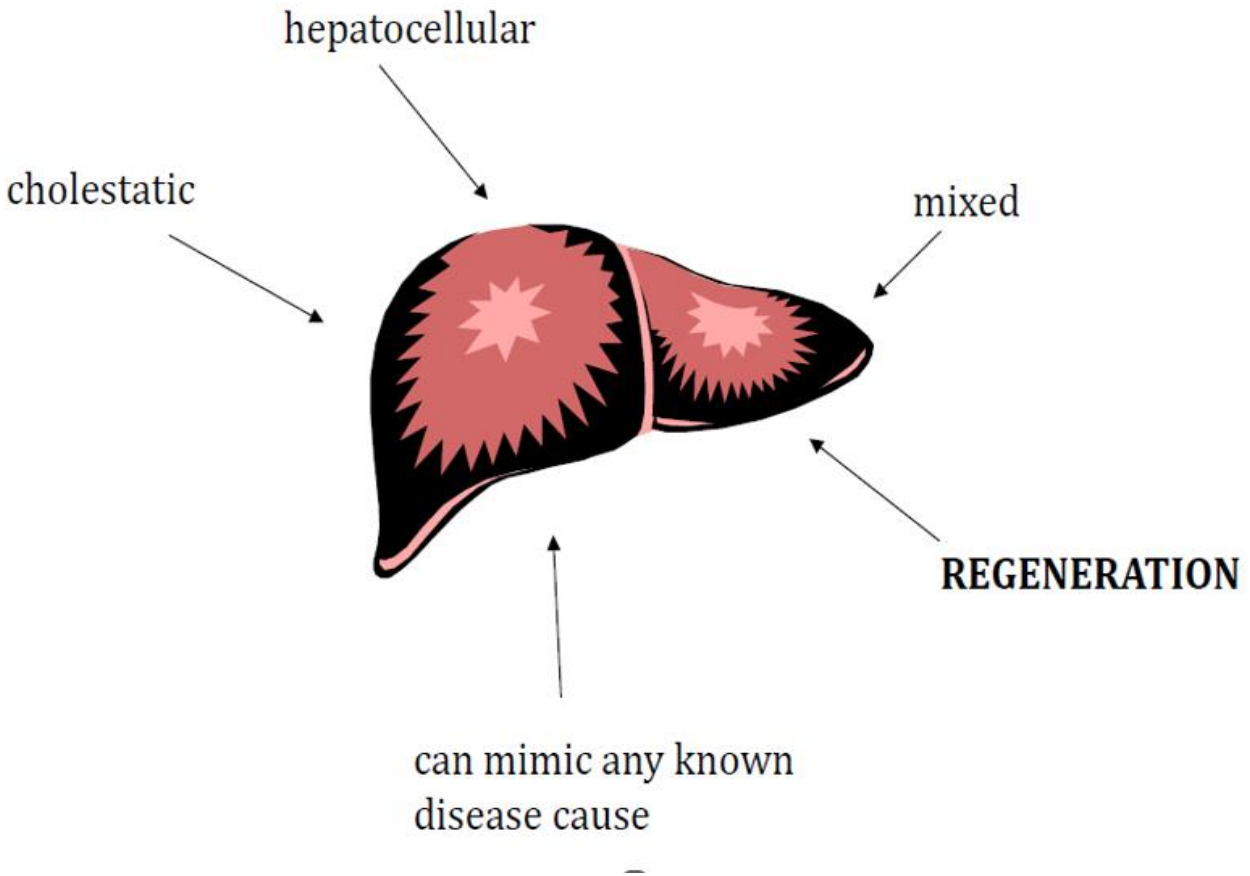
DILI can be classified based on

1. **Clinical presentation:**

- **Hepatocellular**
- **Cholestatic or**
- **Mixed**

2. **Mechanism of hepatotoxicity**

3. Histological appearance from a liver biopsy

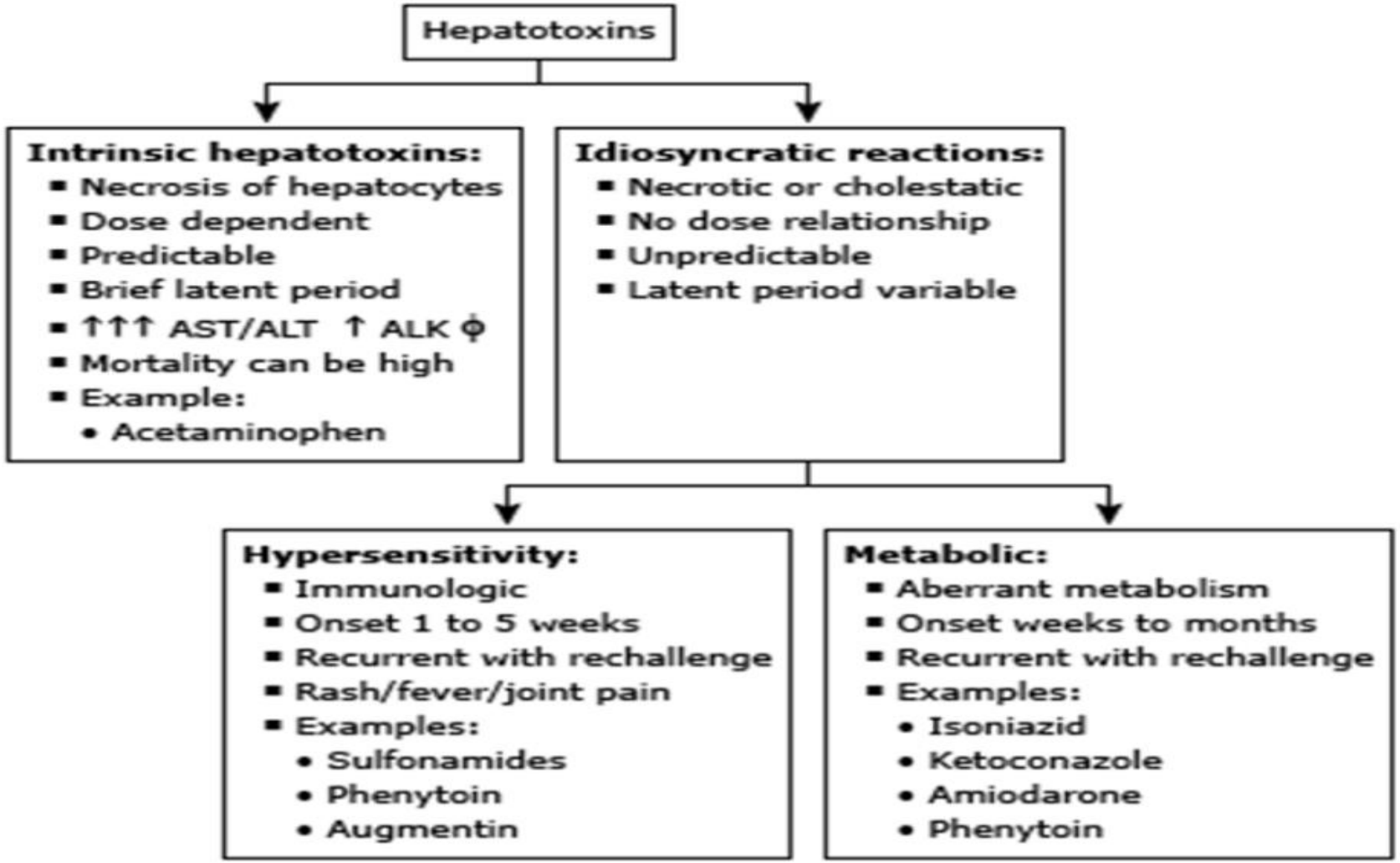


Mechanisms of DILI

- **Two** mechanisms of hepatotoxicity are **intrinsic,** and **idiosyncratic**
- Most cases of DILI are **asymptomatic**; however, the most common sign is **jaundice**.
- **Laboratory tests** in hepatocellular injury will have elevation in aminotransferases,
- while in cholestatic injury, alkaline phosphatase (ALP) is elevated.

- **Treatment begins with the removal of the offending agent, and the prognosis for recovery is usually favorable after discontinuation of the drug.**

Mechanisms of drug-induced liver injury

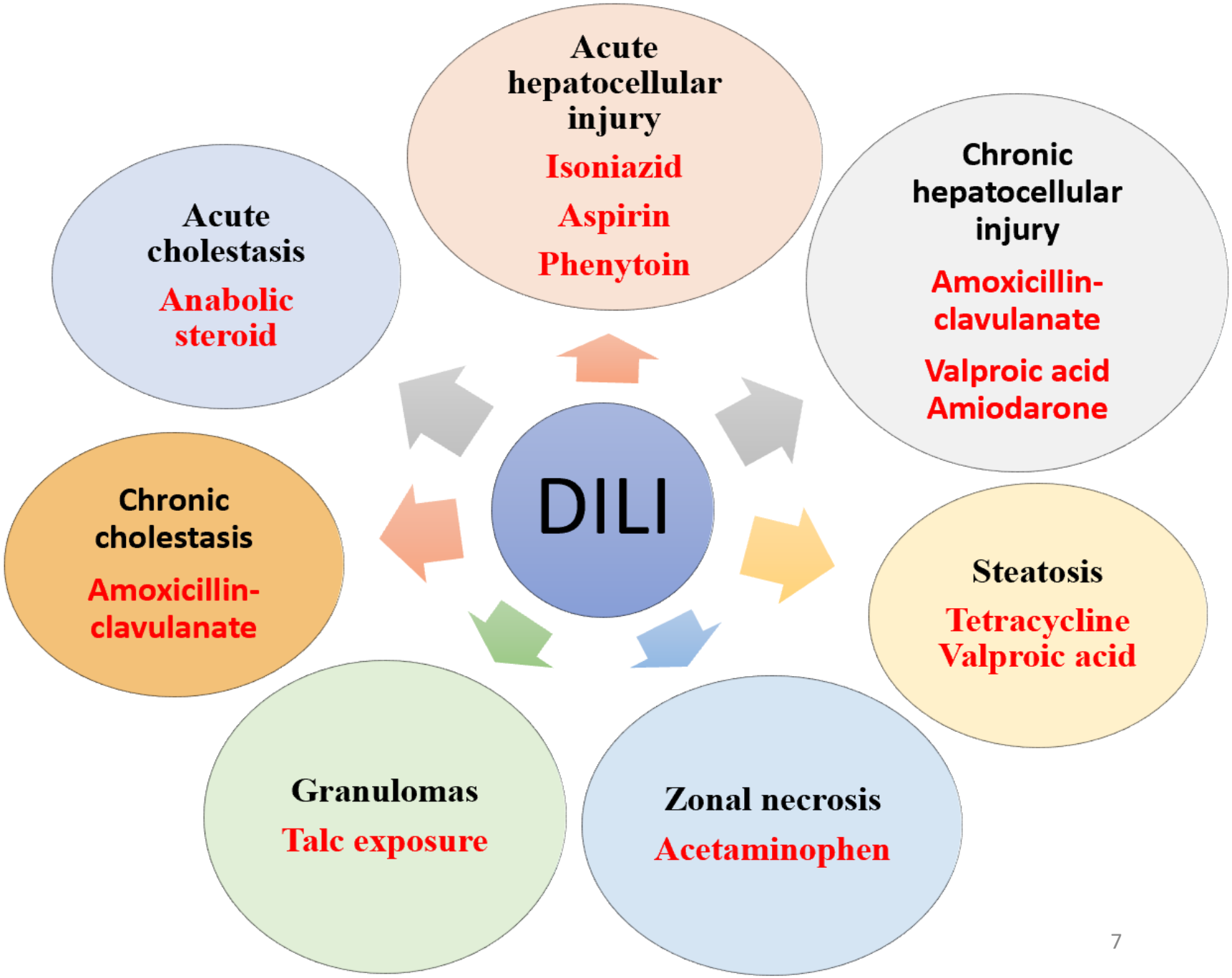


AST: aspartate aminotransferase; ALT: alanine transaminase; ALK: alkaline phosphatase.



DILI

- Examples



Isoniazid-Induced Hepatotoxicity in Tuberculosis Treatment

- A 60-year-old man with tuberculosis presented with malaise, fatigue, jaundice, and right upper quadrant pain after three months of TB treatment.
- Liver enzymes were significantly elevated (ALT: 1,200 U/L, AST: 1,100 U/L, bilirubin: 5 mg/dL).
- The patient had no prior history of liver disease.
- Isoniazid was identified as the cause, and the drug was discontinued.
- Isoniazid ***hepatotoxicity*** is a common complication of antituberculosis therapy that ranges in severity from asymptomatic elevation of serum transaminases to hepatic failure.
- This appears to represent an **idiosyncratic response.**
- INH is metabolized to reactive metabolites mono-acetyl hydrazine, which is further metabolized to a toxic product by cytochrome P450 leading to hepatocellular injury, particularly in patients with genetic susceptibility.

Normal ALT: 7-55 U/L
Normal **AST**: 8-48 U/L
Bilirubin:
Direct: <0.3mg/dL
Total:0.1-1.2 mg/dL

Aspirin-Induced Acute Hepatocellular Injury

- An 8-year-old patient with a history of viral infection (influenza) was on High-dose aspirin (for fever and inflammation).
- The patient presented with confusion, vomiting, and rapid breathing following the use of aspirin for a flu-like illness.
- Laboratory tests showed markedly elevated liver enzymes, including **ALT**: 2,000 U/L (normal:7-55 U/L)and **AST**: 1,800 U/L (normal: 8-48 U/L).
- Blood ammonia levels were elevated, and a liver biopsy showed **microvesicular steatosis**, consistent with **Reye's syndrome**, a rare but serious condition linked to aspirin use during viral infections.

- **Mechanism**: Aspirin can cause mitochondrial dysfunction in the liver, particularly when used in the setting of viral infections, leading to hepatocellular injury and steatosis, as seen in Reye's syndrome.

Phenytoin-Induced Acute Hepatocellular Injury

- **A** 45-year-old female with epilepsy on long-term use Phenytoin medication for seizure management. The patient was admitted with jaundice, fatigue, and nausea. She had been on phenytoin for 6 years, but recently increased her dose due to increased seizure frequency. Laboratory results revealed a **marked elevation in liver enzymes** (ALT: 2,500 U/L, AST: 2,200 U/L), with **bilirubin** of 5.0 mg/dL.

- **Mechanism**: Phenytoin is metabolized by the liver through the **cytochrome P450 system**, and its toxic metabolite can cause **immune-mediated hepatocellular necrosis** and inflammation.

- **Management:** Phenytoin was immediately discontinued, and the patient was started on alternative antiepileptic therapy. Her liver enzymes gradually returned to normal, and her symptoms resolved within several weeks.

Acetaminophen Overdose: Acute Liver Failure

- A 25-year-old woman was admitted to the emergency room after intentionally ingesting a large dose of acetaminophen in a suicide attempt.
- Symptoms included nausea, vomiting, abdominal pain, and jaundice.
- Lab results revealed elevated ALT (10,000 U/L), AST (9,000 U/L), and bilirubin.
- **Mechanism**: Acetaminophen overdose depletes glutathione in the liver, leading to the accumulation of the toxic metabolite called N-acetyl-p-benzoquinone imine (NAPQI), which causes hepatocellular necrosis.
- **Management: N-acetylcysteine, was given:**
- **Replenishes Glutathione:** NAC acts as a precursor to glutathione, helping the body to produce more of it, allowing for the detoxification of NAPQI.
- **Direct Detoxification of NAPQI**
- **Improves Hepatic Blood Flow**
- **Limits Oxidative Stress**

Amoxicillin-Clavulanate-Induced Cholestatic Hepatitis

- A 45-year-old patient developed jaundice, pruritus, and dark urine two weeks after completing a 10-day course of Augmentin for a sinus infection.
- Lab tests showed elevated alkaline phosphatase (ALP), mild elevation in transaminases, and increased bilirubin levels.

AP: 44 to 147 (IU/L)

- A liver biopsy revealed cholestasis. Symptoms persisted for several months, but liver function normalized after discontinuing the medication.
- **Mechanism**: Clavulanic acid is thought to induce an immune-mediated cholestatic liver injury.
- Clavulanic acid trigger an idiosyncratic hypersensitivity reaction in susceptible individuals.
- The immune system may recognize clavulanic acid as foreign and induce an immune response against hepatocytes and cells lining the bile ducts , causing cholestatic liver injury.

Tetracycline-Induced Micro-vesicular Steatosis

- A 40-year-old woman with acne, had been taking tetracycline for several months to treat severe acne. She presented with symptoms of nausea, vomiting, abdominal pain, and fatigue.
- Lab tests revealed elevated liver enzymes (ALT: 600 U/L, AST: 500 U/L), and imaging showed an enlarged liver with fatty infiltration.
- A liver biopsy confirmed microvesicular steatosis, a form of fatty liver disease characterized by the accumulation of small fat droplets within liver cells.
- **Mechanism**: Tetracycline is known to impair mitochondrial function, leading to decreased fatty acid oxidation and accumulation of fat within hepatocytes.
- After discontinuing tetracycline, the patient's symptoms and liver function gradually improved over a period of several weeks.

Valproic Acid-Induced Micro-vesicular Steatosis

- A 17-year-old male had been on valproic acid therapy for epilepsy for two years.
- He presented with symptoms of fatigue, nausea, and weight gain.
- Physical examination revealed hepatomegaly.
- Liver function tests showed elevated transaminases (ALT: 1,200 U/L, AST: 1,100 U/L), and imaging confirmed fatty infiltration of the liver.

- **Mechanism**: Valproic acid can impair mitochondrial β-oxidation of fatty acids, leading to the accumulation of triglycerides in hepatocytes.
- The drug also induces oxidative stress and can cause direct mitochondrial toxicity, contributing to the development of steatosis.
- Valproic acid was discontinued, and the patient's liver function normalized over the course of several months. His seizure management was switched to another antiepileptic drug.

Acute Cholestasis Due to Anabolic Steroid Use

- A 32-year-old male had been using stanozolol for bodybuilding purposes for several months.
- He presented to the hospital with jaundice, dark urine, and pruritus.
- Blood tests revealed significantly elevated bilirubin (12 mg/dL), alkaline phosphatase (ALP: 900 U/L), and gamma-glutamyl transferase (GGT: 800 U/L), while transaminase levels were only mildly elevated.
- Ultrasound ruled out bile duct obstruction. A liver biopsy revealed marked cholestasis.
- **Mechanism**: Anabolic steroids can impair bile flow by affecting bile transporter proteins and causing intrahepatic cholestasis. The cholestatic injury tend to be reversible upon discontinuation of the drug.
- Stanozolol was discontinued, and the patient was treated with ursodeoxycholic acid (UDCA) to improve bile flow. His liver function gradually returned to normal over the next few months, but pruritus persisted for several weeks.

Chronic Hepatocellular Injury from Long-Term Amiodarone Use

- A 68-year-old male had a history of atrial fibrillation and was started on amiodarone therapy three years prior to admission. He had no significant history of alcohol consumption, viral hepatitis, or pre-existing liver disease.
- Liver ultrasound showed hepatomegaly with diffuse fatty infiltration but no evidence of bile duct obstruction.
- Despite discontinuation of amiodarone, the patient's liver function showed only partial recovery. While the ALT and AST levels decreased, the liver fibrosis persisted, indicating chronic liver damage.
- **Mechanism of Injury:**

- **Amiodarone-Induced Hepatotoxicity**: Amiodarone accumulates in the liver and disrupts lipid metabolism, leading to **phospholipidosis** (accumulation of phospholipids within hepatocytes) and **mitochondrial dysfunction**, which can result in **hepatocellular necrosis** and **fibrosis** over time.
- **Chronic Use**: With long-term use, the drug's toxic metabolites accumulate, causing **chronic hepatocellular injury**, often associated with **steatosis**, **inflammation**, and ultimately, **fibrosis** or **cirrhosis** in severe cases.

Non-Steroidal Anti-Inflammatory Drugs (NSAIDs)

- **Sulindac** has been the drug most consistently associated with hepatotoxicity, competitively inhibits **canalicular bile salt transport**, and such inhibition may contribute to ***cholestatic liver injury*** (Liver injury leads to impairment of bile flow presented predominated by itching and jaundice).

Anti-Hyperlipidemic Drugs

- The pattern of liver injury from anti-hyperlipidemics is typically ***hepatocellular or mixed in nature.***
- The proposed mechanisms of hepatotoxicity are varied depending on the drug or drug class, and include effects on the:
- ***Cytochrome P450 system***
- ***Impairment of bile acid*** transport proteins
- ***Immune-mediated inflammatory response*** to the medication or its metabolites, and
- ***Oxidative stress*** due to **intracellular damage**

The antihyperlipidemic drug with the highest potential for hepatic injury is the sustained-release formulation of **Niacin.** The typical pattern of injury involves an elevation in aminotransferase levels although ***a mixed pattern of hepatocellular and cholestatic injury can be seen***

Statins: very rarely cause clinically significant liver injury, although asymptomatic elevation in amino transferases is common.

Anti-Rheumatic Drugs

- ***Methotrexate hepatotoxicity***

Methotrexate as a DMARD, continues to be a first-line drug in the management of early and established RA.

Mechanism: related to the *cellular pathway of the drug*.

Methotrexate therapy in patients with RA has been shown to **raise plasma homocysteine** levels which **generate oxidative stress and activate proinflammatory cytokines**.

The combination of these insults contribute to the activation of **hepatic stellate cells, which leads to liver fibrosis.**

Anti-Rheumatic Drugs

- ***Sulfasalazine hepatotoxicity***
- Most cases occur within the ***first month*** of starting sulfasalazine therapy, and these can present either as a ***hepatocellular or cholestatic pattern of liver injury***.

Anti-Rheumatic Drugs

Azathioprine hepatotoxicity

(Nodular regenerative hyperplasia)

As an immunosuppressant, azathioprine is used in the treatment of a variety of inflammatory disorders including RA.

Hepatotoxic effects associated with azathioprine use include ***nodular regenerative hyperplasia (NRH).***

NRH : may lead to the development of non-cirrhotic portal hypertension

Summary

- mechanisms for drug induced liver injury (DILI) follow a three-step cascade of events:
- Drugs or their metabolites cause cell stress directly or through reactive oxygen species (ROS) during drug oxidation via cytochrome P450.
- impair mitochondrial functions
- trigger immune reactions

References

- **LiverTox: Clinical and Research Information on Drug-Induced Liver Injury.**